

Capstone Project Report

on

ThesisIT: A web-based thesis storage vault

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Chapter 1: Introduction

ThesisIT is a web-based application designed to streamline the management of final-year project records in educational institutions. Traditional methods of storing and managing student projects are inefficient, leading to risks of data loss, limited accessibility, and challenges in collaboration. With the increasing need for digitization in education, ThesisIT aims to provide a centralized, user-friendly, and secure platform for students, faculty, and administrators to store, access, and manage project records efficiently. The platform not only improves project management but also enhances accessibility and reduces paperwork.

Chapter 2: Problem Statement

Managing final year project records is time-consuming, inefficient, and prone to loss or damage, whether in physical or digital form. The lack of a centralized repository limits accessibility and prevents junior students from building on previous work. Hard copies further complicate sharing and collaboration among students and faculty. To address these issues, we are introducing the ThesisIT: Web-Based Thesis Storage Vault, a responsive web application built using AWS, which centralizes project submissions and provides faculty easy access for evaluation.

Chapter 3: Project Overview

ThesisIT is a web-based platform that digitizes and centralizes final-year project management for educational institutions. It enables students to submit, manage, and track their projects while providing faculty with a seamless review and evaluation system. The platform also integrates alumni engagement and user-friendly access features to enhance user experience.

Key Features:

- **Project Submission and Review:** Students can upload reports, while faculty can evaluate and provide feedback.

- **Alumni Engagement:** A dedicated section for alumni to mentor and interact with students.
- **Online Evaluation System:** Faculty members can assess projects digitally, streamlining the grading process.
- **User-Friendly Access:** Intuitive design ensuring seamless navigation for students, faculty, and administrators.
- **Cloud Deployment:** Ensuring scalability, security, and remote accessibility

Chapter 4: Objectives

- To develop a structured repository for final-year project submissions, ensuring systematic storage and retrieval of academic records.
- To enable students to upload, edit, and manage their projects easily, providing an intuitive user interface.
- To provide faculty members with a seamless review and evaluation system, simplifying the grading and feedback process.
- To facilitate alumni engagement by allowing them to share their expertise, guide current students, and provide mentorship.
- To ensure secure storage and easy retrieval of project data with encryption and role-based access control.

Chapter 5: System Design

ThesisIT follows a three-tier architecture comprising:

- **Frontend:** Developed using HTML, CSS, and JavaScript to provide an interactive and responsive user interface.
- **Backend:** Built with Node.js and Express.js to handle authentication, data processing, and API requests.

- **Database:** MongoDB is used for secure and scalable data storage, ensuring efficient project management and retrieval.

User Modules:

- **Student Module:** Allows students to submit projects, track progress, and manage documentation.
- **Faculty Module:** Provides tools for faculty members to evaluate, give feedback, approve or request modifications, and grade student projects.
- **Admin Module:** Manages user roles, monitors submissions, and ensures smooth system operation by handling database operations.
- **Alumni Module:** Enables alumni to engage with students, offer mentorship, share career guidance, and participate in academic discussions.

Chapter 6: Implementation

The project follows an agile development approach with iterative development, regular testing, and continuous feedback integration. The major features implemented include:

- **User Authentication:** Secure login and registration system using JWT (JSON Web Token) authentication to prevent unauthorized access.
- **Project Submission:** Students can upload project reports, source code, and relevant documents while maintaining version history.
- **Evaluation System:** Faculty members can review projects, provide feedback, and assign grades through an intuitive dashboard.
- **Alumni Engagement:** Dedicated space for alumni to interact with students and share insights.
- **User-Friendly Access:** Designed with an intuitive UI to ensure smooth navigation and accessibility for all users.
- **Cloud Deployment:** Hosting on cloud servers for high availability and remote access, ensuring scalability and security.

Chapter 7: Results and Discussions

The initial testing phase of ThesisIT demonstrates significant improvements in accessibility, efficiency, and user engagement. Feedback from students and faculty highlights the platform's ease of use, seamless navigation, and enhanced project submission features. Key performance indicators show a reduction in paperwork, increased project submission rates, and improved faculty-student interaction. Compared to traditional methods, ThesisIT offers a structured and reliable approach to project management, reducing manual effort and ensuring long-term data retention.

Chapter 8: Future Scope

- **Integration of AI-based plagiarism detection** to maintain academic integrity by identifying copied content across projects.
- **Mobile App Development** to provide on-the-go access to project data, notifications, and real-time updates.
- **Blockchain Integration** for secure and immutable project records, ensuring transparency and tamper-proof storage.
- **Automated Report Generation** to assist students in creating project documentation, reducing manual effort.
- **Data Analytics and Insights** to help institutions track project trends, student performance, and faculty evaluations.
- **Integration with Learning Management Systems (LMS)** to streamline project submissions within academic workflows.

Chapter 9: Conclusion

ThesisIT offers a robust solution for project record management, addressing inefficiencies in existing systems. By leveraging modern web technologies, the platform ensures data security, accessibility, and effective project submission and evaluation processes. With future enhancements like AI-based plagiarism detection, blockchain security, and data analytics, ThesisIT has the potential to become a comprehensive platform for academic institutions worldwide.